

Aayush Aashish Mehta

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Professional Summary

- 4+ years of experience in data engineering, AI engineering, cloud engineering, and advanced analytics for enterprise-scale systems and pipelines.
- Expert in building and orchestrating end to end ETL/ELT pipelines using Apache Airflow for batch and incremental processing.
- Strong background in Google Cloud Platform (GCP) tools including BigQuery, Cloud Composer, Cloud Functions, Cloud Storage, DataProc, Pub/Sub, CloudSQL, Vertex AI, and Dialogflow.
- Proficient in developing scalable Customer 360 models by integrating enterprise data from SAP, Salesforce, ServiceNow, and Workday.
- Experienced in migrating SAP HANA calculation views and on-prem SAP datasets to GCP, including authoring fully automated Python DAGs with built-in validation.
- Experienced in building intelligent multi agent chatbots using Google's Agent Development Kit (ADK) at scale.
- Well versed in designing chatbots using Dialogflow for internal automation.
- Designed and maintained text-to-SQL bots enabling non-technical teams to run ad-hoc queries on BigQuery datasets via natural language.
- Delivered real-time and batch data processing solutions, optimizing DAG performance across millions of records.
- Created high-impact dashboards in Looker, Tableau, and Power BI to track KPIs such as engagement scores, marketing ROI, and sales trends.
- Developed Vertex AI pipelines for churn prediction and lead prioritization, integrating outputs with CRM workflows.
- Built and optimized complex SQL transformation logic for use in dashboards and analytics reporting.
- Hands-on experience with NoSQL (MongoDB, Neo4j) and relational databases (PostgreSQL, MySQL, SQL Server, Oracle).
- Experienced in building streaming data apps with Kafka, Pub/Sub, and Cloud Functions.
- Engineered lead enrichment and campaign attribution systems, integrating third-party ad data with internal metrics.
- Deployed anomaly detection tools using Pandas and Sci-kit Learn to monitor weekly revenue and sales deviations.
- Proficient in version control (Git/GitLab), CI/CD workflows, and infrastructure deployments.
- Created schema tracking scripts and validation layers to catch upstream schema changes and ensure data reliability.
- Adept at translating business requirements into scalable and modular cloud-first architecture.
- Documented end-to-end pipelines, transformation logic, and dashboard usage for cross-team onboarding.
- Collaborated with cross-functional teams including product, BI, and DevOps to streamline data workflows.
- Proven ability to enhance operational efficiency by automating legacy reporting workflows and building real-time data systems.

Technical Skills:

Programming Languages:	Python, SQL, shell scripting, C/C++, JavaScript
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Cloud Platforms & Services:	Google Cloud Platform (BigQuery, CloudSQL, Pub/Sub, Cloud Composer, Vertex AI, Cloud Functions, Cloud Storage, Dataproc, Dialogflow, Agent Development Kit), AWS
Data Engineering & Processing:	Apache Airflow, Apache Kafka, Docker, ETL/ELT, PySpark, DBT
Data Analytics & ML:	Pandas, NumPy, Matplotlib, TensorFlow, Keras, PyTorch, Scikit-learn
Databases:	BigQuery, PostgreSQL, MySQL, SQL Server, Oracle, MongoDB, Neo4j, Redshift
BI & Visualization:	Looker, Tableau, Power BI, Seaborn
DevOps & Tooling:	Git, GitLab, CI/CD, Confluence, Jira

PROFESSIONAL EXPERIENCE

ITDataBuild – Flower Mound, TX

Role: Data Engineer (Client: Bayer)

Jan 2026 – Present

- Engineered end-to-end migration pipelines to convert and port all SAP HANA calculation views to Google Cloud Platform (BigQuery), authoring Python-based Airflow DAGs that fully automated extraction, transformation, and loading of view logic and dependencies.
- Implemented comprehensive validation checks within each DAG – covering row counts, schema parity, null distributions, and data-type consistency – to guarantee that migrated calculation views produced results identical to the source SAP HANA views.
- Proposed and designed an improved pipeline architecture to replace the existing migration framework, optimizing task dependencies and removing redundant compute, resulting in meaningful savings on Cloud Composer runtime and BigQuery slot consumption.
- Automated 6 legacy Python jobs that previously required manual triggering by converting them into scheduled Cloud Composer DAGs with retry logic, alerting, and SLA monitoring, eliminating operational toil and improving reliability.
- Developed multiple inbound and outbound data migration DAGs to orchestrate data movement between on-prem SAP systems, Google Cloud Storage, and BigQuery, supporting both batch ingestion and downstream consumer hand-offs.
- Partnered with Bayer’s data and platform teams to translate business requirements into modular, reusable DAG templates that accelerated subsequent migration projects and reduced onboarding time for new datasets.

ITDataBuild – Flower Mound, TX

Role: Data Architect

Jan 2025 – Present

- Created a GCP-based Customer 360 model which involved end to end ETL pipelines integrating data from SAP, ServiceNow, and Salesforce, consolidating cross-platform enterprise data.
- Built Ask-AES, a Vertex AI + Google ADK multi-agent chatbot over BigQuery with inline chart/graph (PNG) rendering and Microsoft Teams integration, enabling conversational enterprise analytics.
- Implemented common analytics flows (Top-N by region/account, 30/90-day trends, account drill-downs) with defaults & validation, reducing ad-hoc SQL queries for routine business questions.

- Computed and standardized 15+ KPIs for Customer 360 and made them queryable in BigQuery for use across finance, sales, and operations.
- Created a natural language Text-to-SQL chatbot layered over the Customer 360 BigQuery tables, allowing internal teams to query KPIs conversationally without SQL knowledge.
- Designed and implemented Looker dashboards for business teams, including a 6-KPI dashboard visualizing historical trends in Account Engagement Scores for global clients.
- Partnered with product and data teams to define KPI logic, metric definitions, and filters for LookML.
- Built a POC for a chatbot using Google's Agent Development Kit (ADK) to automate internal support workflows and enable user-initiated system actions with live agent functionality integrated within Teams and ServiceNow.

Sabre – Dallas, TX

Role: Cloud Data/AI Engineer

May 2025 – Dec 2025

- Created a GCP-based Customer 360 model which involved end to end ETL pipelines integrating data from SAP, ServiceNow, and Salesforce, consolidating cross-platform enterprise data.
- Built Ask-AES, a Vertex AI + Google ADK multi-agent chatbot over BigQuery with inline chart/graph (PNG) rendering and Microsoft Teams integration, enabling conversational enterprise analytics.
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- Built a POC for a chatbot using Google's Agent Development Kit (ADK) to automate internal support workflows and enable user-initiated system actions with live agent functionality integrated within Teams and ServiceNow.
- Maintained and expanded a suite of modular Apache Airflow DAGs for scheduled incremental batch processing and transformation of Salesforce data into BigQuery.
- Migrated Airflow SQL to dbt (BigQuery); added layered models, freshness/tests, Slim CI on PRs, and docs/exposures tied to Looker.
- Developed reusable DAG templates to support ad-hoc project needs, e.g., fetching and mapping metadata for Salesforce account table migration.
- Wrote Python scripts for schema validation, duplicate checks, and transformation consistency during ETL jobs.

Analytics IT –

Role: Cloud Data/AI Engineer

Sept 2022 – June 2023

- Designed and deployed automated ETL pipelines using Apache Airflow and BigQuery to support customer segmentation initiatives for a global retail client.
- Built a datalake/lakehouse by integrating data from HR systems, Salesforce, and ServiceNow; consolidated 18+ business KPIs and developed interactive dashboards to drive cross-functional insights and reporting.

- Standardized transformations in dbt (BigQuery): built staged/intermediate/marts layers for HR/Salesforce/ServiceNow, incremental campaign/event models partitioned by event_date, and added freshness + data tests (unique/not_null/relationships).
- Built a Dialogflow-based NLP chatbot for a lead enrichment platform, allowing marketing teams to access campaign metrics and client profiles via voice or text queries.
- Created an ML pipeline using Vertex AI to predict churn probabilities, integrating the output with Salesforce CRM for personalized retention workflows.
- Developed a KPI tracking dashboard in Looker to visualize campaign ROI, customer funnel health, and drop-off patterns across cohorts.
- Engineered modular DAGs for batch ingestion of campaign metadata and event logs from Google Ads into BigQuery, used for attribution modeling.
- Participated in the architecture and delivery of a product intelligence system to aggregate feature usage metrics across app versions.
- Tuned SQL transformation logic in BigQuery to improve dashboard load times by 35% and reduce query costs.
- Led the development of a proof-of-concept for a real-time pricing insights engine using Pub/Sub and Cloud Functions.
- Designed schema monitoring scripts to track upstream schema changes and auto-notify analysts on pipeline breakage risks.
- Collaborated with cross-functional teams (BI, DevOps, Product) to define KPIs and automate report generation.

Nividata Consultancy –

Role: Data Engineer

May 2021 – June 2022

- Led the development of batch ETL pipelines using Python and Apache Airflow to process product and sales data for an e-commerce client.
- Integrated AWS S3 as a centralized staging layer and built data ingestion workflows for MySQL and MongoDB sources.
- Developed a data validation layer using Python and Pandera to enforce schema integrity before loading into Redshift.
- Built interactive Tableau dashboards tracking sales performance, conversion rates, and product return ratios across regions.
- Collaborated with business analysts to define KPI metrics and optimize SQL logic for reporting dashboards.
- Implemented Python scripts to detect and alert anomalies in weekly sales trends, improving revenue monitoring responsiveness.
- Participated in migrating legacy Excel-based reporting workflows to automated dashboards, cutting manual effort by over 50%.
- Documented all ETL processes and transformation logic in Confluence, enabling team-wide transparency and maintainability.

Bosky Buildcon –

Role: Data Analyst

May 2020 – Dec 2020

- Conducted cost-efficiency analysis across 12 active construction projects using SQL and Excel, identifying a 15% overhead savings opportunity through material reallocation.
- Built interactive visual reports using Matplotlib and Seaborn to track project burn rates, workforce allocation, and vendor performance.

- Automated daily extraction of site data from internal tools using Python, reducing manual reporting workload by 40%.
- Created Tableau dashboards to help site supervisors monitor budget utilization and forecast material demand.
- Collaborated with the operations team to define key project KPIs and built custom SQL views for each.
- Standardized data entry workflows and implemented validation scripts to reduce reporting inconsistencies across project sites.
- Documented all analysis workflows and developed reporting templates for future analysts to replicate.

ACTIVITIES

Youth India Foundation –

Role: Head of Finance

Nov 2021 – Nov 2022

- Managed budgeting, expense tracking, and revenue planning for state and national events, ensuring financial transparency and accountability.
- Used Excel to analyze event profitability, visualize revenue trends, and prepare weekly financial reports for leadership.
- Coordinated intern recruitment and onboarding while leading weekly meetings to align teams on tasks and operational goals.
- Collaborated with event and operations teams, using data insights to optimize resource allocation and reduce overall event costs.

TwoWaits –

Role: Web Developer

Nov 2021 – Dec 2021

- Developed a secure email authentication module in Node.js and integrated it with a Flutter client application for seamless user onboarding.
- Contributed to frontend improvements in the Flutter app, including enhancing UI responsiveness, refining input validation, and ensuring smoother user flows during login and signup.
- Contributed to documentation of API endpoints and authentication workflows for future developer reference.
- Supported QA testing by identifying bugs, replicating issues, and validating fixes across different user flows.

PUBLICATION

DeSculpt: Indian Temple Sculpture Iconography

- Co-authored and published the above, proposing a deep learning framework for 9-class classification of Navagraha deity sculptures.
- Addressed extreme data scarcity by applying extensive data augmentation to improve generalization.
- Applied transfer learning on pre-trained CNN architectures (ResNet50, VGG16, MobileNet) and conducted systematic architecture comparison.
- Experimented with hybrid CNN feature extraction + SVM classifiers for comparative performance evaluation.
- Achieved 91% classification accuracy, demonstrating deep learning applicability in cultural heritage digitization.
- Documented dataset creation methodology, preprocessing pipeline, architecture comparison, and evaluation metrics in the published paper.

EDUCATION

Bachelor of Technology in Computer Science Engineering – S.R.M. University, Chennai, India

CGPA: 9.38/10

Master of Science in Computer Science – Arizona State University, USA

CGPA: 3.71/4

Awarded NAMU Scholarship